



COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF ELECTRONICS

PROJECT REPORT **Sobel Edge Detector**

Submitted by

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design to deploy- short term Course

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Abstract

This project presents the design and implementation of a Sobel Edge Detection Processor using Verilog HDL and a complete RTL-to-GDSII design flow. Edge detection is a fundamental operation in digital image processing and computer vision applications, where the Sobel operator is widely employed for detecting object boundaries and intensity variations in images. The processor operates on a 3×3 pixel window and computes horizontal and vertical image gradients to generate an edge intensity output.

The functionality of the Sobel processor was verified through simulation using a SystemVerilog testbench, and functional coverage analysis was performed to ensure adequate verification of various image patterns and edge detection scenarios. The synthesized design was generated using Cadence Genus, followed by Logic Equivalence Checking (LEC) to verify functional equivalence between the RTL description and the synthesized gate-level netlist.

Physical implementation was carried out using Cadence Innovus, including floorplaning, placement, Clock Tree Synthesis (CTS), routing, and optimization. Static Timing Analysis (STA) was performed using Cadence Tempus to validate the timing requirements and ensure successful design signoff. The final design achieved successful timing closure, zero routing overflow, and satisfied all design rule constraints.

This project demonstrates the complete ASIC design methodology from RTL design to physical implementation and timing signoff using industry-standard EDA tools. The implemented Sobel Edge Detection Processor serves as a practical example of image processing hardware acceleration and provides valuable exposure to modern VLSI design and verification methodologies.

1. Introduction

The Sobel Edge Detection Processor is a hardware-based image processing system designed to detect edges in digital images by computing horizontal and vertical intensity gradients. In this project, a Sobel processor is implemented using Verilog HDL, where a 3×3 pixel window is processed to calculate the gradient components (G_x and G_y) and generate an edge intensity output. The hardware accelerator is designed to provide efficient and high-speed edge detection suitable for real-time image processing applications.

The project follows a complete ASIC design flow, starting from RTL design and functional verification, followed by synthesis, Logic Equivalence Checking (LEC), floor-planning, placement, Clock Tree Synthesis (CTS), routing, and Static Timing Analysis (STA). Industry-standard Cadence EDA tools are used throughout the implementation process.

The objective of this project is to demonstrate the design, verification, and physical implementation of an image processing hardware accelerator while achieving timing closure and generating a manufacturable GDSII layout through a complete RTL-to-GDSII implementation flow.

2. Objectives

- To design and implement a Sobel Edge Detection Processor using Verilog HDL.
- To develop a hardware accelerator capable of detecting edges in digital images by computing horizontal and vertical image gradients.
- To verify the functionality of the design through simulation and coverage analysis.
- To perform logic synthesis and generate an optimized gate-level netlist.
- To validate functional equivalence between the RTL and synthesized netlist using Logic Equivalence Checking (LEC).
- To implement the complete physical design flow, including floorplanning, placement, Clock Tree Synthesis (CTS), and routing.
- To perform Static Timing Analysis (STA) and achieve timing closure with positive setup and hold timing margins.
- To generate a final GDSII layout demonstrating successful RTL-to-GDSII implementation of the Sobel Edge Detection Processor.
- To gain practical exposure to industry-standard ASIC design methodologies and Cadence EDA tools used in modern VLSI implementation flows.

3. Project Plan

- Phase 1: Requirement Analysis
 - Define the functionality of the Digital Voting Machine.
 - Identify inputs, outputs, and operational requirements for five candidates.
- Phase 2: RTL Design
 - Develop the Verilog HDL code for the voting machine.
 - Implement vote counting and reset functionality.
- Phase 3: Functional Verification
 - Create a SystemVerilog testbench.
 - Simulate the design and verify correct vote counting for all candidates.
 - Perform coverage analysis to ensure adequate test coverage.
- Phase 4: Logic Synthesis
 - Synthesize the RTL design using Cadence Genus.
 - Generate gate-level netlist and analyze area, power, and timing reports.
- Phase 5: Logic Equivalence Checking (LEC)
 - Verify functional equivalence between RTL and synthesized netlist.
- Phase 6: Physical Design
 - Perform floorplanning and power planning.
 - Execute placement and Clock Tree Synthesis (CTS).

- Complete routing and physical optimization.
- Phase 7: Static Timing Analysis
- Perform setup and hold timing analysis using Cadence Tempus.
- Resolve timing violations and achieve timing closure.
- Phase 8: Final Verification and GDSII Generation
- Validate the final routed design.
- Generate the GDSII layout for fabrication readiness.
- Document results and prepare the project report.

4. Images and Report

RTL-to-GDSII Implementation of Sobel Edge Detection Accelerator

This section presents the complete RTL-to-GDSII implementation of the Sobel Edge Detection Accelerator using Cadence EDA Tools. The design is initially developed at the Register Transfer Level (RTL) and verified through simulation. Subsequently, the complete ASIC design flow, including synthesis, floorplanning, power planning, placement, clock tree synthesis (CTS), routing, timing analysis, and physical verification, is performed to achieve a GDSII-ready implementation. Post-route timing analysis, Static Timing Analysis (STA), gate-level simulation, and Logical Equivalence Checking (LEC) are conducted to ensure functional correctness and timing closure.

RTL Design

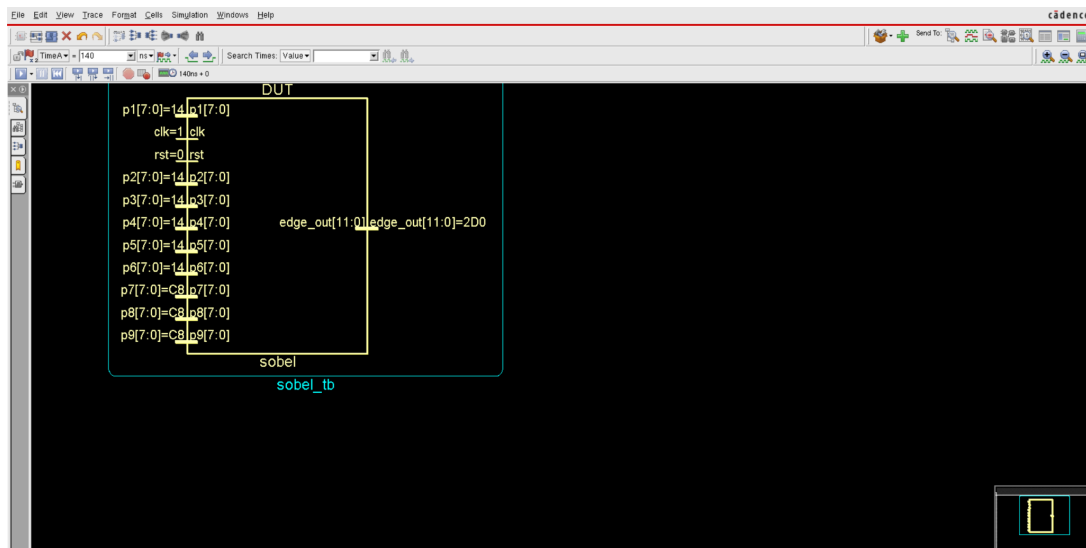


Figure 4.1: RTL Design – schematic

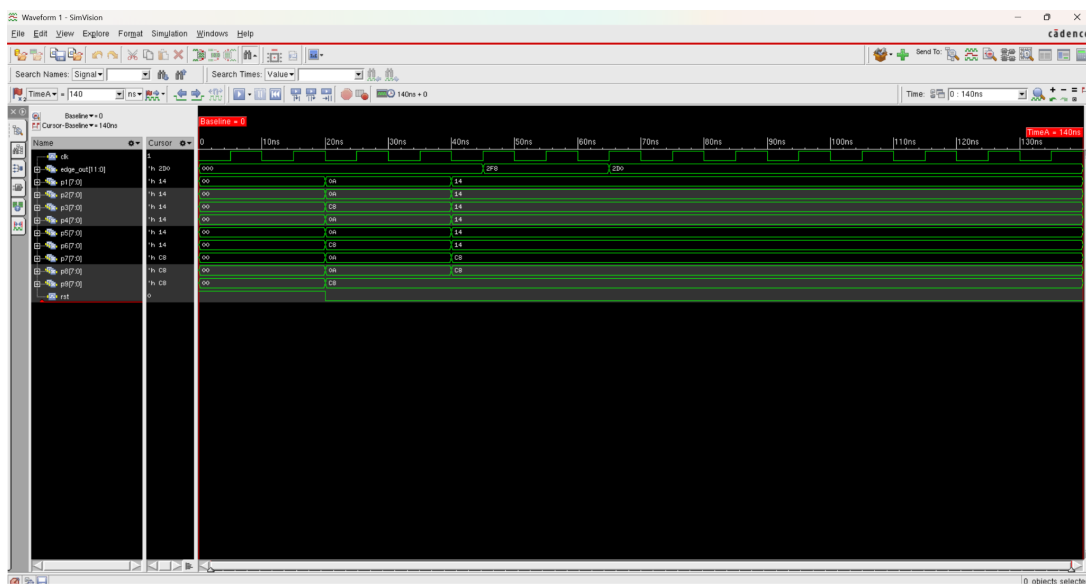


Figure 4.2: RTL Design – Simulation 2

Coverage

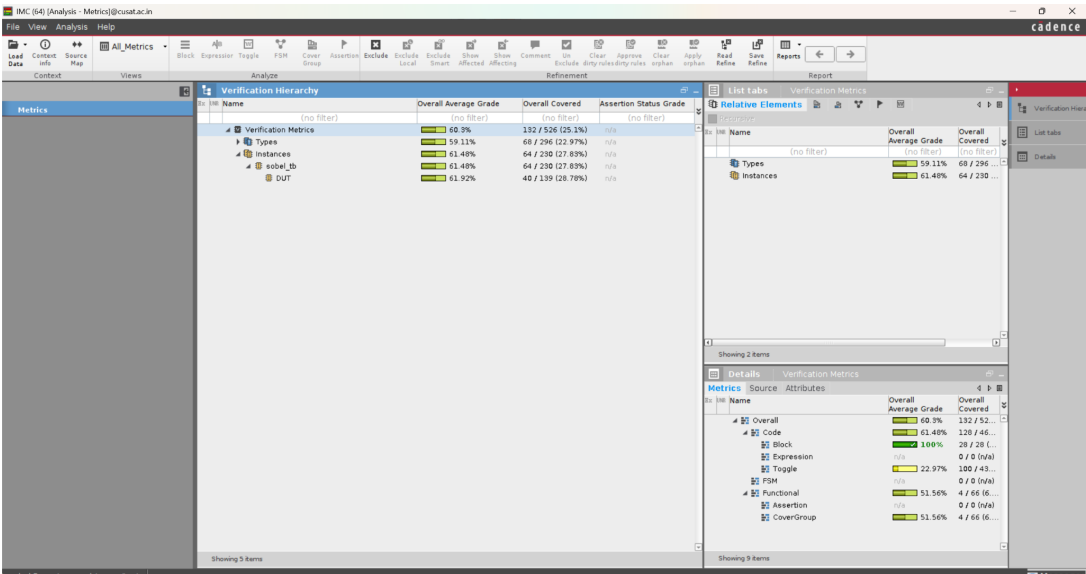


Figure 4.3: Coverage Report

Synthesis

Area

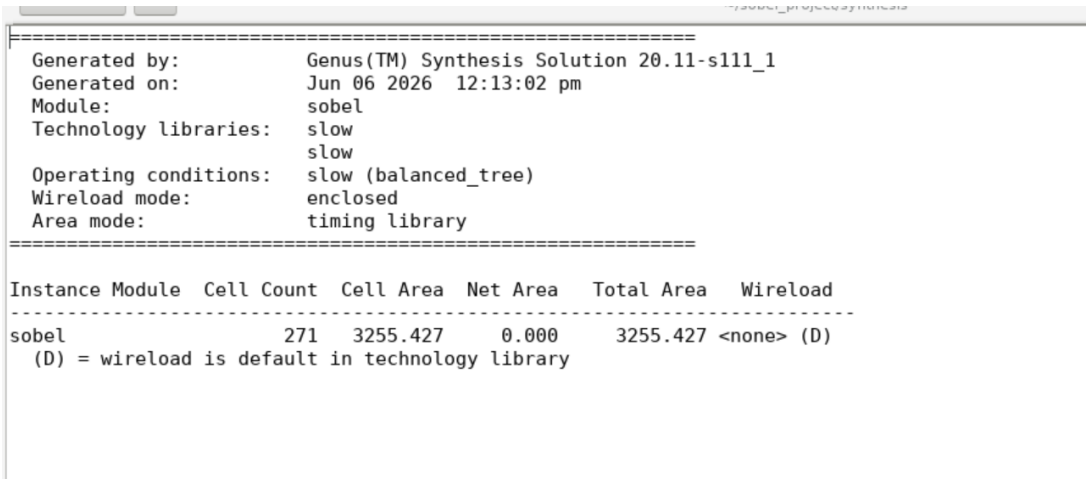


Figure 4.4: Synthesis – Area Report

Timing and Power

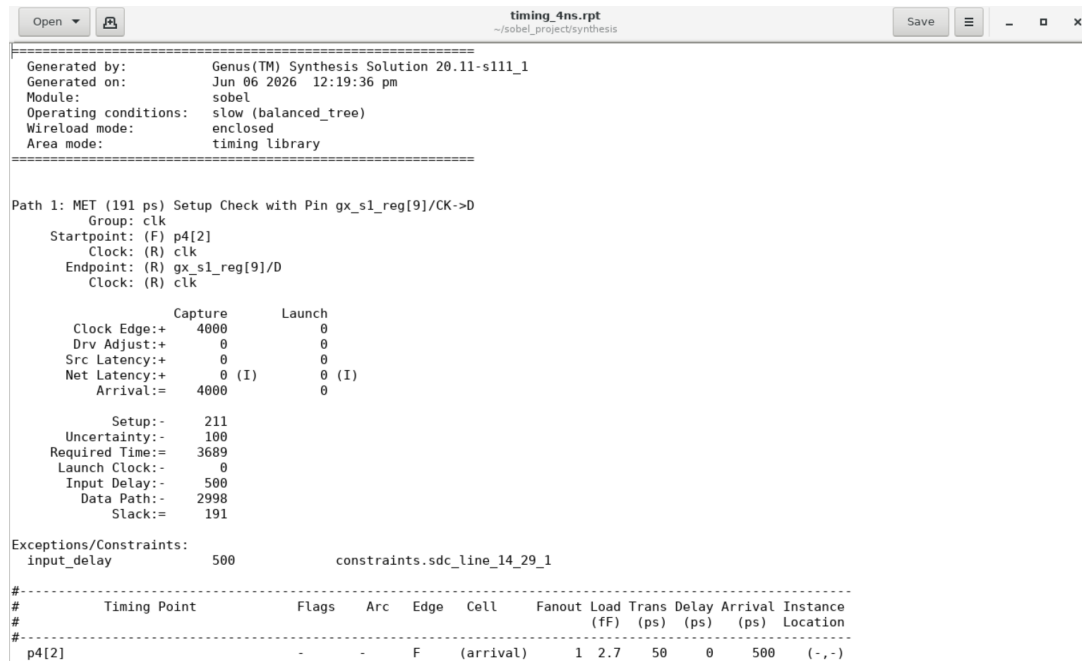


Figure 4.5: Synthesis – Timing Report

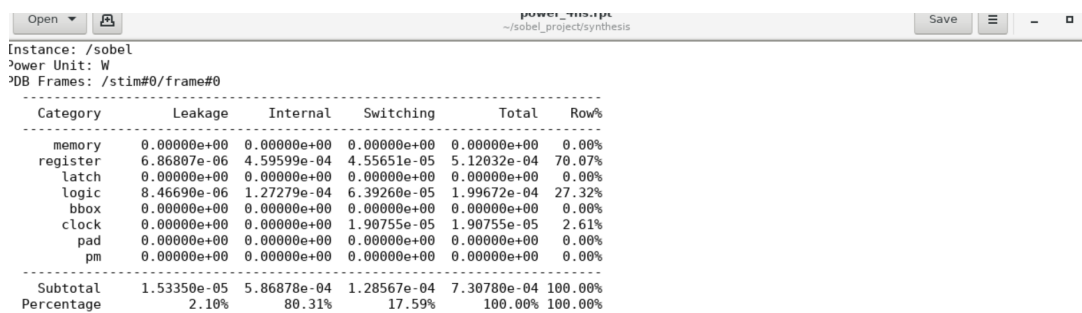


Figure 4.6: Synthesis – Power Report

LEC

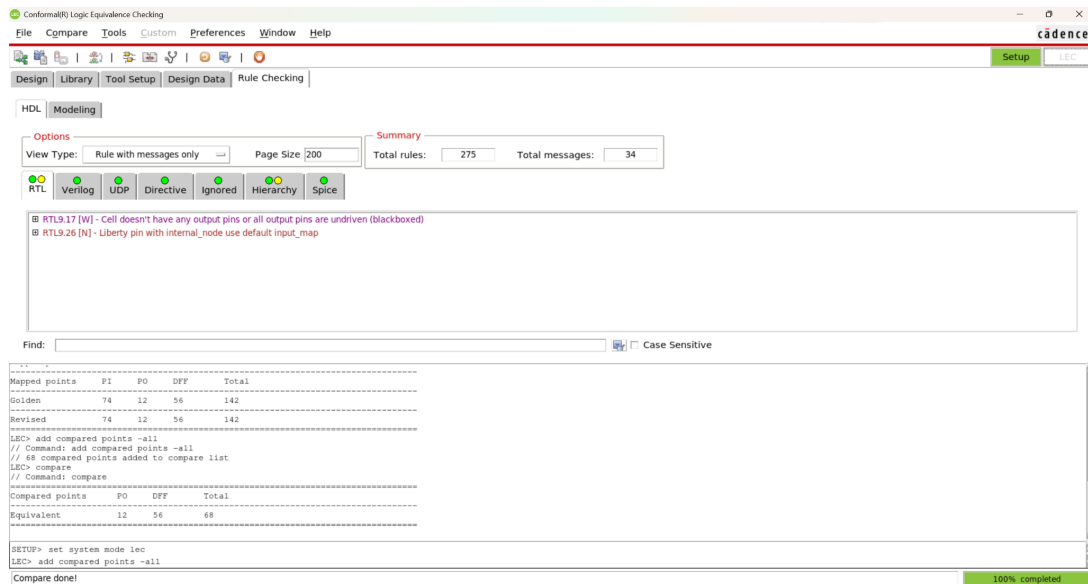


Figure 4.7: LEC – Result 1

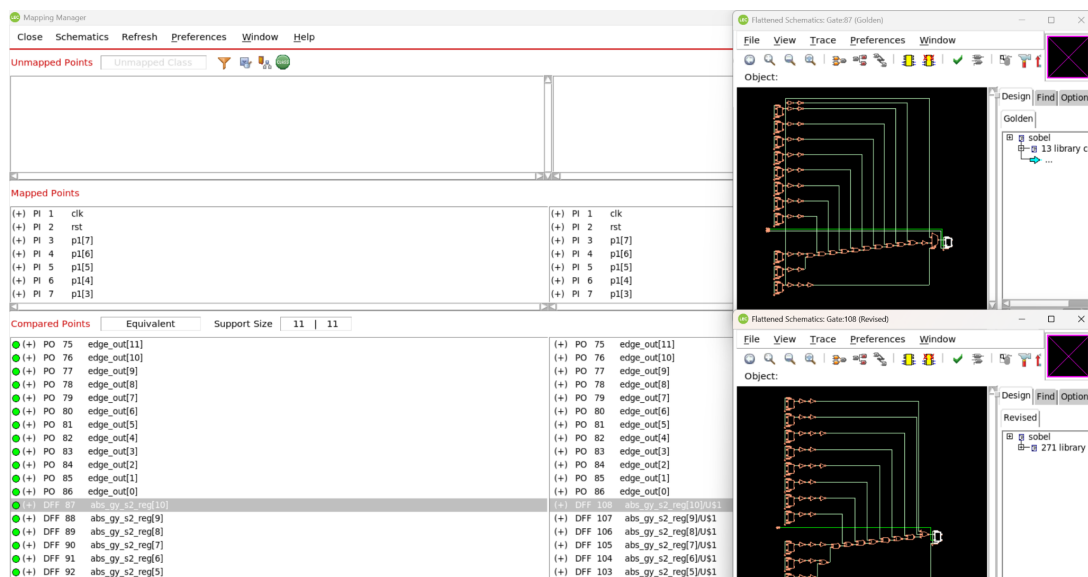
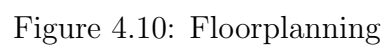


Figure 4.8: LEC – Result 2

Floorplanning



Pre-CTS

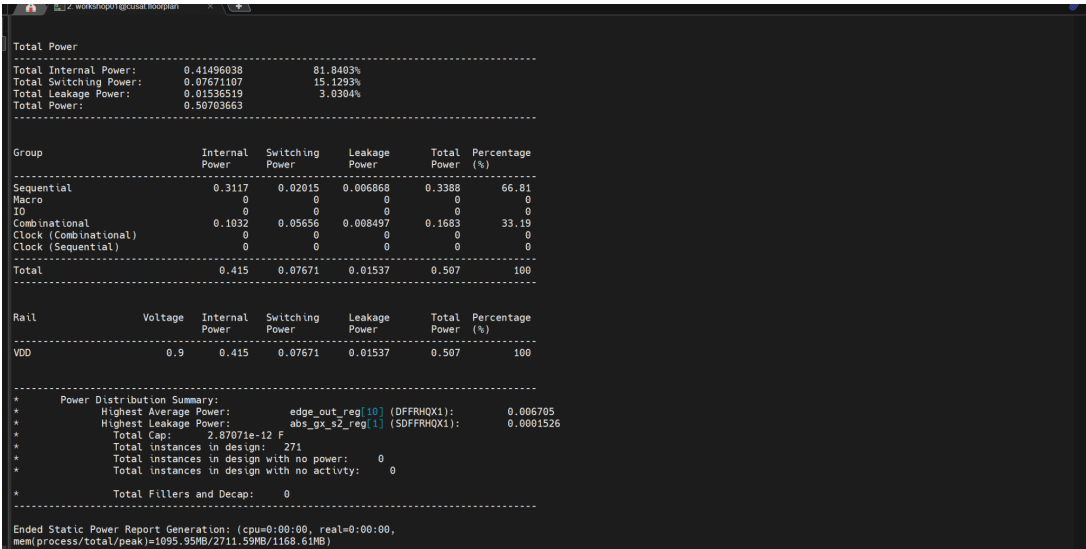


Figure 4.11: Pre-CTS – Result 1

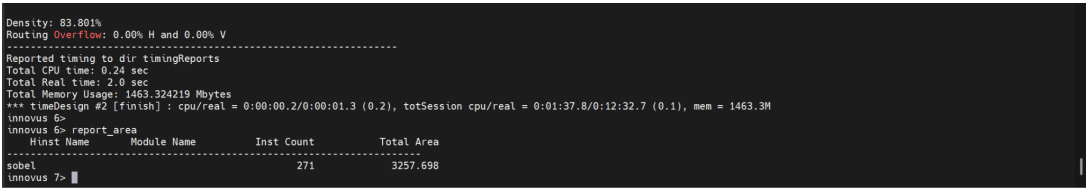


Figure 4.12: Pre-CTS – Result 2

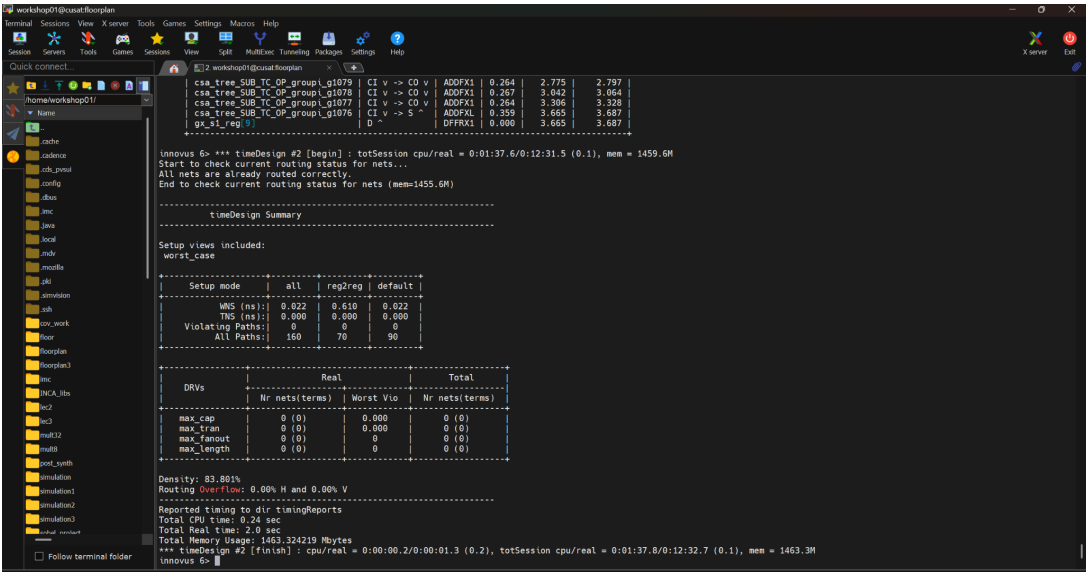


Figure 4.13: Pre-CTS – Result 3

Clock Tree Synthesis

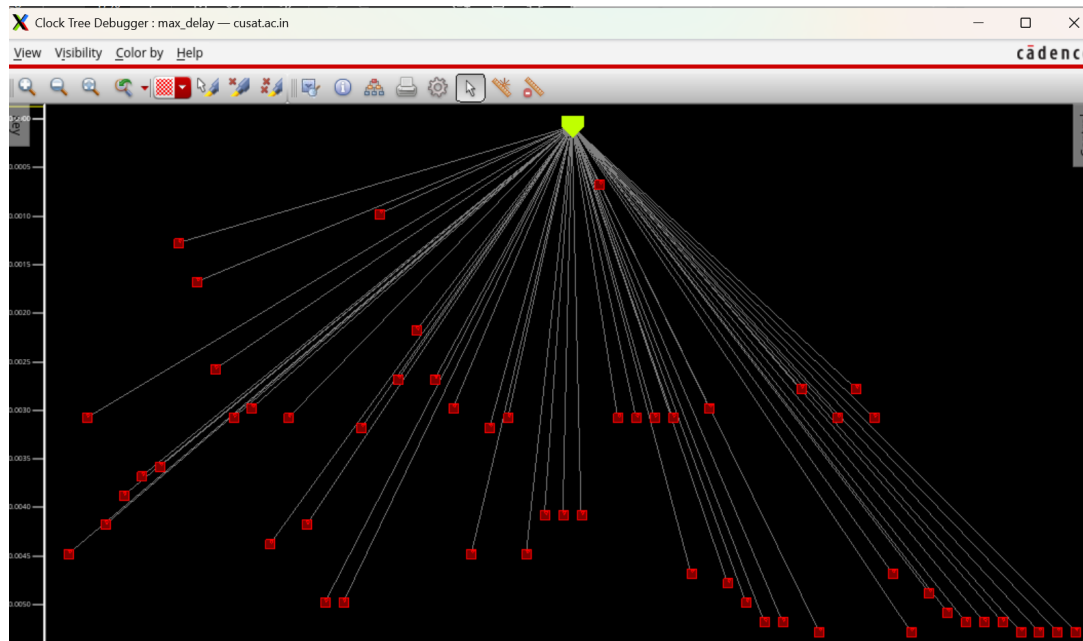


Figure 4.14: Clock Tree Synthesis

Post-CTS Report

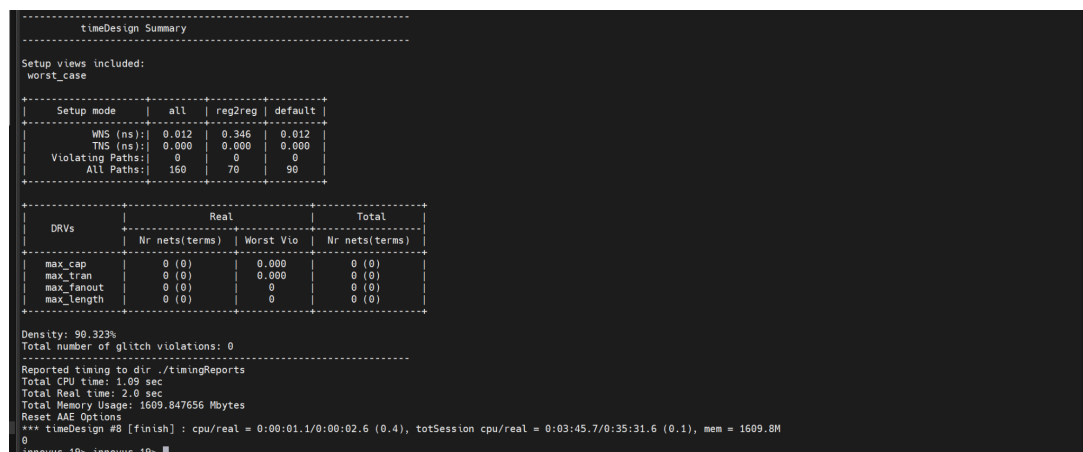


Figure 4.15: Post-CTS Report – Result 1

```

-----
timeDesign Summary
-----
Hold views included:
best_case

-----
| Hold mode | all | reg2reg | default |
-----
| WNS (ns): | 0.005 | 0.005 | 0.372 |
| TNS (ns): | 0.000 | 0.000 | 0.000 |
| Violating Paths: | 0 | 0 | 0 |
| All Paths: | 160 | 70 | 90 |
-----

Density: 90.323%

-----
Reported timing to dir ./timingReports
Total CPU time: 1.21 sec
Total Real time: 1.0 sec
Total Memory Usage: 1541.730469 Mbytes
Reset AAE Options
*** timeDesign #7 [finish] : cpu/real = 0:00:01.2/0:00:01.5 (0.8), totSession cpu/real = 0:03:37.6/0:33:47.4 (0.1), mem = 1541.7M
0

```

Figure 4.16: Post-CTS Report – Result 2

```

# Analysis Mode: MMCM Non-ULV
# Parasitics Mode: No SPEF/RCDB
# Signoff Settings: SJ Off
=====
Calculate delays in BcWc mode...
Start delay calculation (fullDC) (1 T). (MEM=1476.62)
Total number of fetched objects 421
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
End delay calculation. (MEM=1490.3 CPU=0:00:00.1 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=1490.3 CPU=0:00:00.1 REAL=0:00:00.0)
*** Done Building Timing Graph (cpu=0:00:00.1 real=0:00:00.0 totSessionCpu=0:02:31 mem=1490.3M)

-----
timeDesign Summary
-----
Setup views included:
worst_case

-----
| Setup mode | all | reg2reg | default |
-----
| WNS (ns): | 0.022 | 0.610 | 0.022 |
| TNS (ns): | 0.000 | 0.000 | 0.000 |
| Violating Paths: | 0 | 0 | 0 |
| All Paths: | 160 | 70 | 90 |
-----

-----
| DRVs | | Real | | Total |
|-----|-----|-----|-----|
| | | Nr nets(terms) | Worst Vio | Nr nets(terms) |
|-----|-----|-----|-----|
| max_cap | 0 (0) | 0.000 | 0 (0) |
| max_tran | 0 (0) | 0.000 | 0 (0) |
| max_fanout | 0 (0) | 0 | 0 (0) |
| max_length | 0 (0) | 0 | 0 (0) |
|-----|-----|-----|-----|

Density: 83.801%
Routing Overflow: 0.00% H and 0.00% V

-----
Reported timing to dir timingReports
Total CPU time: 0.42 sec
Total Real time: 1.0 sec
Total Memory Usage: 1459.566406 Mbytes
*** timeDesign #3 [finish] : cpu/real = 0:00:00.4/0:00:01.4 (0.3), totSession cpu/real = 0:02:31.6/0:25:29.3 (0.1), mem = 1459.6M
innovus >

```

Figure 4.17: Post-CTS Report – Result 3

Post-Route Report

```

-----
timeDesign Summary
-----
Hold views included:
best_case

-----
| Hold mode | all | reg2reg | default |
-----
| WNS (ns): | 0.005 | 0.005 | 0.372 |
| TNS (ns): | 0.000 | 0.000 | 0.000 |
| Violating Paths: | 0 | 0 | 0 |
| All Paths: | 160 | 70 | 90 |
-----

Density: 90.323%

-----
Reported timing to dir timingReports
Total CPU time: 1.11 sec
Total Real time: 1.0 sec
Total Memory Usage: 1556.199219 Mbytes
Reset AAE Options
*** timeDesign #10 [finish] : cpu/real = 0:00:01.1/0:00:01.4 (0.8), totSession cpu/real = 0:04:15.5/0:41:08.4 (0.1), mem = 1556.2M
innovus 21>

```

Figure 4.18: Post-Route Report – Result 1

```
-----
timeDesign Summary
-----

Setup views included:
worst_case

-----
| Setup mode | all | reg2reg | default |
-----
| WNS (ns): | 0.012 | 0.344 | 0.012 |
| TNS (ns): | 0.000 | 0.000 | 0.000 |
| Violating Paths: | 0 | 0 | 0 |
| All Paths: | 160 | 70 | 90 |
-----

-----
| DRVs | Real | Total | |
|---|---|---|---|
| | Nr nets(terms) | Worst Vio | Nr nets(terms) |
|-----|-----|-----|
| max_cap | 0 (0) | 0.000 | 0 (0) |
| max_tran | 0 (0) | 0.000 | 0 (0) |
| max_fanout | 0 (0) | 0 | 0 (0) |
| max_length | 0 (0) | 0 | 0 (0) |
|-----|-----|-----|

Density: 90.323%
Total number of glitch violations: 0
-----
Reported timing to dlr timingReports
Total CPU time: 1.17 sec
Total Real time: 3.0 sec
Total Memory Usage: 1619.917969 Mbytes
Reset AAE Options
*** timeDesign #9 [finish] : cpu/real = 0:00:01.2/0:00:02.6 (0.5), totSession cpu/real = 0:04:10.6/0:40:09.7 (0.1), mem = 1619.9M
innovus 21> █
```

Figure 4.19: Post-Route Report – Result 2

Post-Route LEC

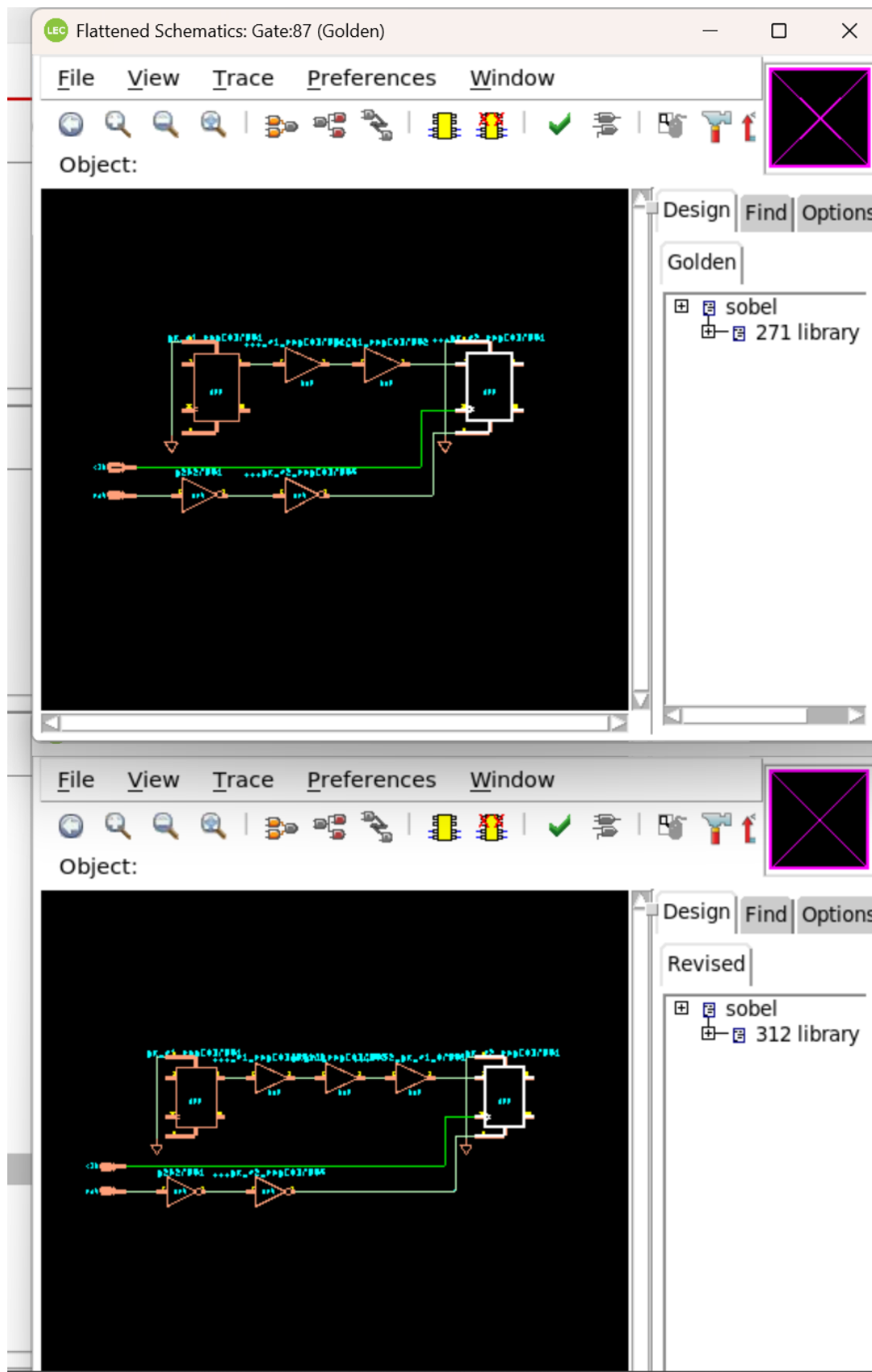


Figure 4.20: Post-Route LEC

The screenshot shows the Post-Route LEC tool interface. It has a menu bar with 'Close', 'Schematics', 'Retresh', 'Preferences', 'Window', and 'Help'. Below the menu bar are three main sections: 'Unmapped Points', 'Mapped Points', and 'Compared Points'.

Unmapped Points: This section is currently empty.

Mapped Points: This section displays two columns of points. The left column lists points from PI 1 to PI 7, and the right column lists points from PI 1 to PI 7. The points are: PI 1 clk, PI 2 rst, PI 3 p1[7], PI 4 p1[6], PI 5 p1[5], PI 6 p1[4], and PI 7 p1[3].

Compared Points: This section displays two columns of points. The left column lists points from PO 75 to PO 92, and the right column lists points from PO 75 to PO 92. The points are: PO 75 edge_out[11], PO 76 edge_out[10], PO 77 edge_out[9], PO 78 edge_out[8], PO 79 edge_out[7], PO 80 edge_out[6], PO 81 edge_out[5], PO 82 edge_out[4], PO 83 edge_out[3], PO 84 edge_out[2], PO 85 edge_out[1], PO 86 edge_out[0], DFF 87 abs_gx_s2_reg[0]/U51, DFF 88 abs_gx_s2_reg[1]/U51, DFF 89 abs_gx_s2_reg[2]/U51, DFF 90 abs_gx_s2_reg[3]/U51, DFF 91 abs_gx_s2_reg[4]/U51, and DFF 92 abs_gx_s2_reg[5]/U51.

Figure 4.21: Post-Route LEC – Result 2

The screenshot shows the Post-Route LEC tool interface with the 'Summary' tab selected. The 'Options' section shows 'View Type: Rule with messages only' and 'Page Size: 200'. The 'Summary' section shows 'Total rules: 275' and 'Total messages: 34'. The 'Results' section shows a list of messages, including 'RTL9.17 [W] - Cell doesn't have any output pins or all output pins are undriven (blackboxed)' and 'RTL9.26 [N] - Liberty pin with internal_node use default input_map'.

Find: ☐ Case Sensitive

Results:

Mapped points	P1	PO	DFF	Total
Golden	74	12	56	142
Revised	74	12	56	142

LEC> add compared points -all
 // Command: add compared points -all
 // 48 compared points added to compare list
 LEC> compare
 // Command: compare

Compared points	PO	DFF	Total
Equivalent	12	56	68

SETUP> set system mode lec
 LEC> add compared points -all
 Compare done!

100% completed

Figure 4.22: Post-Route LEC – Result 3

Figure 4.24: SDF Delay Simulation

Static Timing Analysis

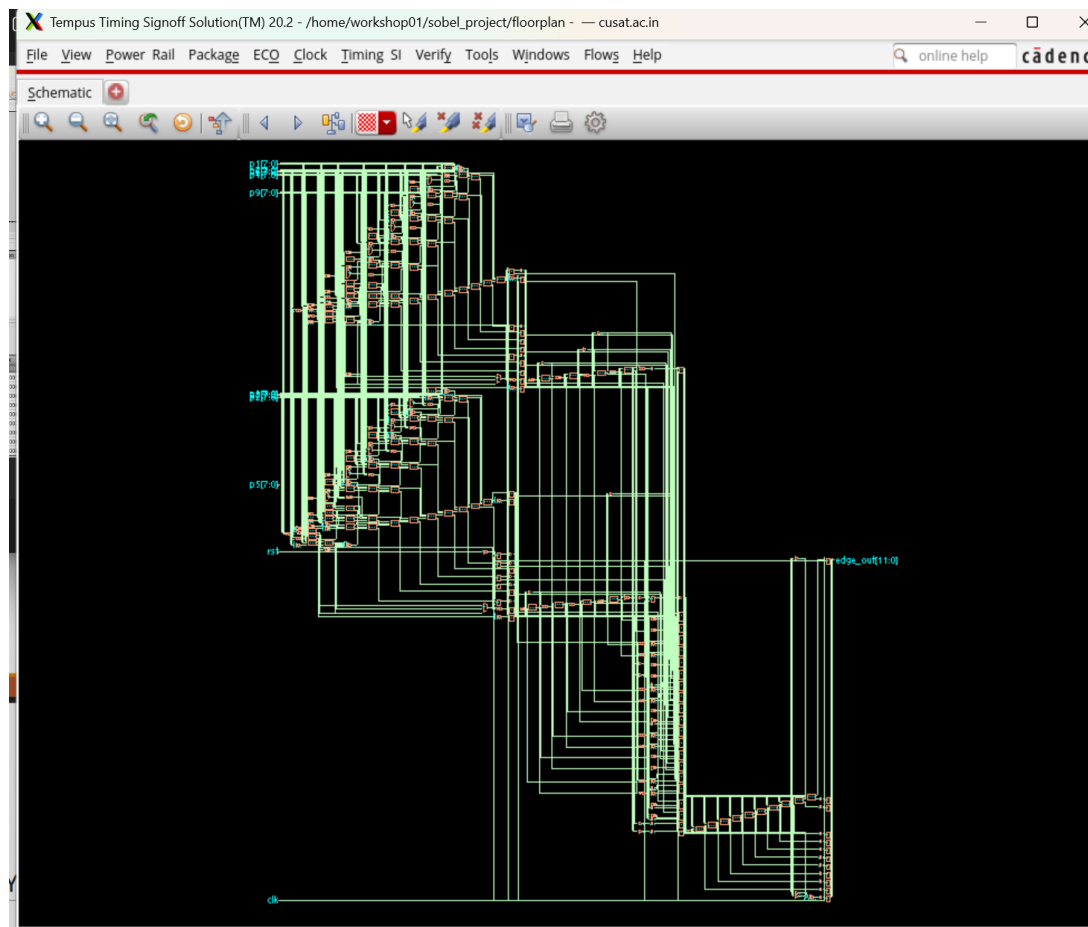


Figure 4.25: Static Timing Analysis – Result 1

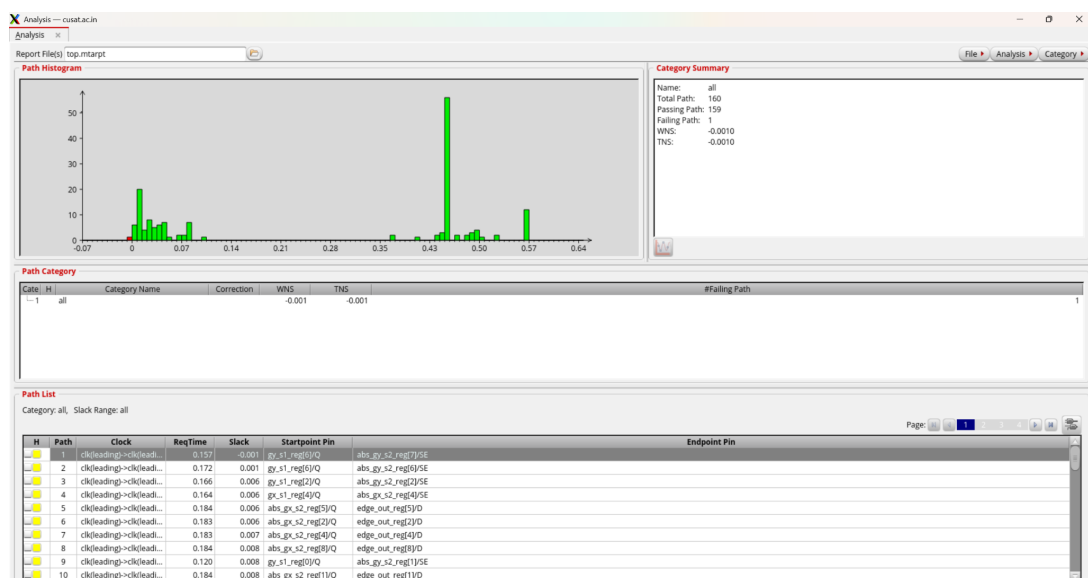


Figure 4.26: Static Timing Analysis – Result 2

Timing Analysis Observation

Post-route timing analysis was performed using both Cadence Innovus and Cadence Tempus. The Innovus timing reports indicated positive setup and hold slacks, suggesting that the design met the specified timing constraints during implementation. However, signoff analysis in Cadence Tempus reported a marginal hold violation with a Worst Negative Slack (WNS) of -0.001 ns (1 ps) in the best-case hold corner.

The discrepancy between Innovus and Tempus results can be attributed to differences in timing analysis methodologies, extraction accuracy, and signoff-level pessimism applied by Tempus. Since Tempus performs more accurate signoff timing analysis using detailed parasitic information and advanced variation modeling, the slight negative slack reported by Tempus is considered the final signoff result.

The observed violation is extremely small (1 ps) and can be regarded as a marginal hold issue arising from signoff pessimism rather than a significant functional timing failure. Nevertheless, in an industrial ASIC design flow, the Tempus signoff report is treated as the authoritative timing result, and such violations would typically require additional hold optimization before tape-out.